



BOARD OF STUDIES

2025

Undergraduate Programs

Institute of Computing,
Kohat University of Science & Technology



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1. INTRODUCTION

The Institute of Computing (IoC), formerly known as Institute of Information Technology (IIT), is one of the oldest and largest teaching institute at Kohat University of Science and Technology (KUST). The institute offers various undergraduate and graduate degrees, certification programs in addition to provides trainings and services to other departments at the campus. The undergraduate programs offered are Bachelor of Science in Computer Science and Bachelor of Science in Software Engineering. While Master of Science in Computer Science, Master of Science in Data Science (MSDS), Master of Science in Software Engineering (MSSE) and Ph.D. in Computer Science are the graduate programs. Additionally, the institute provides a variety of short-term training and instructional programs, including the Prime Minister's Kamyab Jawan Program and Cisco Networking Training.

All the degree programs are on par with the Higher Education Commission (HEC), National Computing Education Accreditation Council (NCEAC) and other universities of international repute. Apart from offering regular academic programs, the department organizes seminars, workshops, and conferences on the latest emerging topics and issues in computing on regular basis. The department enjoys the support of an outstanding team of highly qualified, experienced, and young faculty members. The existing faculty members are well qualified from reputed national and international universities. The faculty provides expertise in various specialization areas such as IT Management, Security Technologies, Network Technologies, Software Technologies, Artificial Intelligence, Data Mining, Mobile Technologies, Distributed Systems and Theoretical Computer Sciences. IoC also offers state-of-the-art facilities that include an auditorium, well-equipped lecture rooms, computing Labs, final year projects Lab, Research and Development Cell, Huawei Lab, and Web Development Cell. The latest teaching aids such as multimedia projectors, LEDs and OHPs are essential tools for students and teachers. The building of IoC is purpose-built and well equipped with all such types of essential teaching aids.

The department educates and conducts research covering wide areas of attractive educational programs for students to learn from the basics to advanced technologies related to computer science and information technology. Students are taught and groomed to become leading developers and researchers who are highly motivated and have practical, creative and management skills to drive an advanced next-generation information society in all industrial zones.

1.1. VISION AND MISSION

a. Vision Statement

To become a center of excellence for producing top-notch computer science professionals with strong computing and research skills, leadership abilities, and moral principles to meet the demands of industry and society.

b. Mission Statement

The mission of Institute of Computing is to provide student with top-notch education in computing along with cutting-edge practical skills and intrapreneurship training so that they may play a significant role in the society as well as industry through critical thinking, innovation, and research.

1.2. ACADEMIC RESOURCES

Resources	Objectives	Major Equipment
3 Computing Labs	These computing labs are used for lab work of various courses.	Computer systems with network connectivity
Classrooms	10 classrooms used by undergraduate and graduate students	Equipped with LEDs
Conference Hall	Used for seminars, presentations and other similar activities	Multimedia projector and sound system
Other facilities	Girls common room, staff meeting room, washrooms, stores	None
Incubation Center	Used for student startups and industrial linkages	Computer systems with network connectivity

1.3. FACULTY

S#	Name	Designation
1.	Dr. Amjad Mehmood	Director/ Associate Professor
2.	Dr. Shafiullah Khan	Professor (on Leave)
3.	Dr. Muhammad Hammad Hassan	Assistant Professor
4.	Dr. Jawad Ashraf	Assistant Professor (On Leave)
5.	Dr. Zeeshan Iqbal	Assistant Professor
6.	Dr. Abdul Shahid	Assistant Professor (On Leave)
7.	Dr. Saima Hassan	Assistant Professor
8.	Dr. Muhammad Altaf Khan	Assistant Professor
9.	Dr. Muhammad Irfan Uddin	Assistant Professor
10.	Dr. Shafaq Mussadiq	Assistant Professor
11.	Dr. Rabia Khan	Assistant Professor
12.	Dr. Noor Mast	Lecturer
13.	Dr. Muhammad Adnan	Lecturer
14.	Dr. Muhammad Muneer Umar	Lecturer
15.	Dr. Sana Ullah Khan	Lecturer
16.	Dr. Muhammad Roman	Lecturer (On Leave)
17.	Mr. Abdur Rehman	Lecturer
18.	Mr. Muhammad Abrar Khan	Lecturer

19.	Mr. Qadeem Khan	Lecturer
20.	Mr. Faisal Baseer	Lecturer
21.	Mr. Tahir Naeem	Lecturer
22.	Mr. Muhammad Ali Zeb	Lecturer
23.	Mr. Ilyas Ahmad	Lecturer
24.	Mr. Sajid Ullah	Lecturer
25.	Dr. Samina Amin	Lecturer (Contract)
26.	Mr. Umar Ayaz	Lecturer (Contract)
27.	Mr. Abdullah	Lecturer (Contract)

1.4. ACADEMIC PROGRAMS WITH ANNUAL INTAKES

Program	Yearly Intake
Bachelor of Science in Computer Science (BSCS)	100(open) + 100 (self)
Bachelor of Science in Software Engineering (BSSE)	50
Bachelor of Science in Cyber Security (BSCYS)	No Intake till 2025
Bachelor of Science in Artificial Intelligence (BSAI)	No Intake till 2025
Bachelor of Science in Data Science (BSDS)	No Intake till 2025
Master of Science in Computer Science (MSCS)	Varies
Ph.D. in Computer Science (Ph.D.(CS))	Varies

2. UNDERGRADUATE COURSES

2.1. COURSE CATEGORIZATION AND DISTRIBUTION

Areas	NCEAC/HEC Proposed		KUST IoC BS Program	
	Credit Hours	Courses	Credit Hours	Courses
 General Education	30	12	31	12
 Mathematics & supporting	12	4	12	4
 Elective Supporting	3	1	3	1
 Computing Core	46	14	46	14
 Domain Core	18	6	18	6
 Domain Elective	21	7	21	7
Totals	130	44	131	44
<i>Additional HEC Required General Education courses</i>				
 Understanding of Quran - I	1	1	1	1
 Understanding of Quran - II	1	1	1	1
 Pakistan Studies	2	1	3	1
 Internship	0	1	0	1
Total Program Credits	134	48	136	48

2.2. DEFINITION OF COURSE CODES

Course Category	Course Code Pattern
General Education Courses	Given by parent departments
Mathematics & supporting	Given by parent departments
Social Sciences	Given by parent departments
Computing Core Courses	CC###
Computer Science Program Courses	CS###
Software Engineering Program Courses	SE###

Cyber Security Program Courses	CY###
Artificial Intelligence Program Courses	AI###
Data Science Program Courses	DS###
Post Graduate Diploma in CS	PGD###

Each code consists of a two-letter prefix followed by a three-digit number (###). The codes apply to general, core, and program-specific courses. Program course codes cover both domain and elective courses. The three digits in the course code represent:

- The first digit: Academic level/year of the course (1–4)
- The second digit: Course category within the program (e.g., Programming, Systems, AI, etc.)
- The third digit: Sequence number of the course within that category and level

Courses in General Education, Mathematics, and Social Sciences are assigned codes by their respective departments.

2.3. DOMAINS OF COMPUTING COURSES

Category Digit	Computing Core (CC)	CS Program (CS)	SE Program (SE)	CY Program (CY)	AI Program (AI)	DS Program (DS)
0	Fundamentals	Fundamentals	Fundamentals	Fundamentals	Fundamentals	Fundamentals
1	Programming & Computing	Programming & Computing	Programming & Computing	Programming & Computing	Programming & Computing	Programming & Computing
2	Data/ Information Management / Database	Data/ Information Management / Database	Software Process & Modeling	System & Network Security	Machine Learning	Data Handling & Preprocessing
3	Computer Sys/ Hardware/ OS/ Electronics	Computer Sys/ Hardware/ OS/ Electronics	Software Design & Architecture/ Construction	Cryptography & Secure Systems	Deep Learning	Data Mining & Warehousing
4	Multimedia / Web	Multimedia / Web	Multimedia / Web	Digital Forensics & Compliance	Computer Vision & NLP	Big Data & Cloud
5	Software Engineering/ Artificial Intelligence	Software Engineering/ Artificial Intelligence	Quality, Testing, Maintenance	Risk Mgmt. & Governance/ Incident Response	Robotics, Agents & Planning	Statistical Modelling
6	Network/ Communication Systems	Network/ Communication Systems	Project Mgmt/ Communication Systems	Cybersecurity Ethics & Society/ Communication Systems	AI Ethics & Society/ Communication Systems	Visualization & Analytics / Communication Systems
7	Elective/ Other	Elective/ Other	Elective/ Other	Elective/ Other	Elective/ Other	Elective/ Other
8	Elective/ Other	Elective/ Other	Elective/ Other	Elective/ Other	Elective/ Other	Elective/ Other
9	Project	Project	Project	Project	Project	Project

2.4. GENERAL EDUCATION COURSES

S#	Code	Course Title	Sub-Category	Credit Hrs	Pre-Req
1	CS101	Applications of Information and Communication Technologies	ICT	3(2+1)	
2	ENG151	Functional English	ENG	3(3+0)	
3	ENG152	Expository Writing	ENG	3(3+0)	
4	MATH101	Quantitative Reasoning – I	QR	3(3+0)	
5	MATH102	Quantitative Reasoning – II	QR	3(3+0)	
6	RS101	Islamic Studies	-	3(3+0)	
7	PS106	Ideology and Constitution of Pakistan	-	2(2+0)	
8	BS121	Principles of Management	Social Sciences	2(2+0)	
9	PHY101 PHY104 ES101	<i>(Choose only ONE Course)</i> -Introductory Mechanics -Introductory Electricity -Everyday Science	Natural Sciences	3(3+0)	
10	PH101 JMC102	<i>(Choose only ONE Course)</i> Introduction to Philosophy Introduction to Journalism <i>(Any other Arts & Humanities Course)</i>	Arts & Humanities	2(2+0)	
11	SWS102	Social Ethics and Civic Engagement	Ethics	2(2+0)	
12	BS311	Entrepreneurship	-	2(2+0)	
13	PS101	Pakistan Studies	-	3(3+0)	
14	---	Understanding of Quran - I	-	1(0+1)	
15	---	Understanding of Quran - II	-	1(0+1)	

2.5. MATHEMATICS AND SUPPORTING COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req.
	<i>Choose any ONE Courses</i>			
1	ENG254 ENG334	Advanced Academic Reading and Writing Technical and Business Writing	3(3+0)	
	<i>Choose any THREE Courses</i>			
2	MATH103	Discrete Mathematics	3(3+0)	
3	MATH212	Linear Algebra	3(3+0)	
4	STAT221	Basic Inferential Statistics	3(3+0)	
5	MATH104	Calculus-I		
6	MATH272	Ordinary Differential Equations		
7	MATH471	Mathematical Modeling		
8	STAT102	Introduction to Probability distributions		

2.6. ELECTIVE SUPPORTING COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req
	<i>Choose only ONE Course</i>			
1	BS261 PSY101 ECON121 ECON111	Financial Accounting Introduction to Psychology Principles of Macroeconomics Principles of Microeconomics	3(3+0)	

	SWS101	Introduction to sociology		
	BS141	Introduction to marketing		

2.7. COMPUTING CORE COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req
1.	CC102	Programming Fundamentals	4 (3+1)	
2.	CC111	Object Oriented Programming	4 (3+1)	CC102 (PF)
3.	CC131	Digital Logic Design	3 (2+1)	
4.	CC211	Data Structures	4 (3+1)	CC102(PF)
5.	CC221	Database Systems	4 (3+1)	
6.	CC231	Computer Organization and Assembly Language	3 (2+1)	CC131(DLD)
7.	CC232	Operating Systems	3 (3+0)	
8.	CC251	Software Engineering	3 (3+0)	
9.	CC261	Computer Networks	3 (2+1)	
10.	CC252	Artificial Intelligence	3 (2+1)	
11.	CC262	Information Security	3 (3+0)	
12.	CC411	Analysis of Algorithms	3 (3+0)	CC211 (DS)
13.	CC498	Final Year Project - I	3 (0+3)	
14.	CC499	Final Year Project - II	3 (0+3)	CC498 (FYP-I)
15.	CC497	Internship	0	

2.8. BSCS PROGRAM DOMAIN AND ELECTIVE COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req.
Domain Core Courses (Select All)				
1.	CS311	Theory of Automata	3 (3+0)	
2.	CS315	Compiler Construction	3 (3+0)	
3.	CS321	Advanced Database Management Systems	3 (2+1)	CC221(DB)
4.	CS331	Computer Architecture	3 (3+0)	CC131(DLD)
5.	CS342	HCI and Computer Graphics	3 (3+0)	
6.	CS461	Parallel and Distributed Computing	3 (3+0)	
Domain Electives (Choose SEVEN Courses)				
1.	CS371	Web Development	3 (2+1)	
2.	CS372	Advanced Programming	3 (2+1)	CC111(OOP)
3.	CS373	Mobile Application Development	3(2+1)	
4.	CS374	Game Design and Development	3(2+1)	CC102(PF)
5.	CS375	Software Testing and Quality Assurance	3(3+0)	CC251(SWE)
6.	CS376	Advanced Web Development	3(2+1)	CS371(WD)
7.	AI322	Machine Learning	3(3+0)	
8.	CS378	Wireless Networks	3(3+0)	CC261(CN)
9.	CS379	Search Engine Optimization	3 (2+1)	
10.	CS381	Human Computer Interaction	3 (3+0)	
11.	CS382	Computer Graphics	3 (2+1)	
12.	CS471	Distributed Wireless Networks	3(3+0)	

13.	CS472	Internet of Things and Applications	3 (3+0)	CC261(CN)
14.	CS473	Sensor Networks and Applications	3(3+0)	CC261(CN)
15.	CS474	Artificial Neural Networks	3(3+0)	CC351(AI)
16.	CS475	Recommender Systems	3(3+0)	CC351(AI)
17.	CS476	Blockchain Technologies	3(3+0)	
18.	CS477	Object Oriented Analysis and Design	3(3+0)	
19.	<i>Any other programs' elective or domain course subject to the approval of Advisor/Director</i>			

2.9. BSSE PROGRAM DOMAIN AND ELECTIVE COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req.
Domain Core Courses (Select All)				
1.	SE321	Software Requirement Engineering	3 (3+0)	
2.	SE331	Software Construction and Development	3 (3+0)	CC251(SWE)
3.	SE332	Software Design and Architecture	3 (3+0)	CC251(SWE)
4.	SE351	Software Quality Engineering	3 (3+0)	CC251(SWE)
5.	SE461	Software Project Management	3 (3+0)	
6.	CS461	Parallel and Distributed Computing	3 (3+0)	
Domain Electives (Choose SEVEN Courses)				
1.	CS375	Software Testing and Quality Assurance	3(3+0)	CC251(SWE)
2.	SE372	Agile Software Development	3 (3+0)	
3.	SE373	Usability Engineering	3 (3+0)	
4.	SE374	Software Metrics	3(3+0)	CC251(SWE)
5.	SE375	Fundamentals of UX		
6.	SE376	Introduction to User-Centered Design		
7.	SE471	Software Verification and Validation	3 (3+0)	CC251(SWE)
8.	SE472	Software Engineering Economics	3 (3+0)	
9.	SE473	Software Re-Engineering	3(3+0)	
10.	<i>Any other programs' elective or domain course subject to the approval of Advisor/Director</i>			

2.10. BSAI PROGRAM DOMAIN AND ELECTIVE COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req.
Domain Core Courses (Select All)				
1.	AI311	Programming for AI	3 (2+1)	CC102(PF)
2.	AI322	Machine Learning	3 (2+1)	
3.	AI331	Artificial Neural Networks & Deep Learning	3 (2+1)	
4.	AI351	Knowledge Representation & Reasoning	3 (2+1)	
5.	AI441	Computer Vision	3 (2+1)	
6.	CS461	Parallel & Distributed Computing	3 (3+0)	
Domain Electives (Choose SEVEN Courses)				
1.	AI371	Mathematics for AI and Data Science	3 (3+0)	QR1
2.	DS331	Data Science	3(2+1)	
3.	DS333	Data Mining	3(2+1)	CC221(DB)
4.	AI374	Natural Language Processing	3(2+1)	
5.	AI375	Business Intelligence and Decision Support Systems	3 (3+0)	

6.	AI376	Knowledge Based Systems	3 (3+0)	
7.	AI377	Fuzzy Systems	3 (3+0)	QR-2
8.	AI378	Fundamentals of Large Language Models	3 (3+0)	
9.	AI379	Digital Image Processing	3 (2+1)	
10.	AI471	Reinforcement Learning	3 (3+0)	AI322(ML)
11.	AI472	Swarm Intelligence	3 (3+0)	CC351 (AI)
12.	AI473	Agent Based Modeling	3 (3+0)	CC351(AI)
13.	<i>Any other programs' elective or domain course subject to the approval of Advisor/Director</i>			

2.11. BSCYS PROGRAM DOMAIN AND ELECTIVE COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req.
Domain Core Courses (Select All)				
1.	CY321	Cyber Security	3 (3+0)	
2.	CY322	Network Security	3 (3+0)	CC261(CN)
3.	CY331	Secure Software Design and Development	3 (3+0)	
4.	CY351	Information Assurance	3 (3+0)	
5.	CY441	Digital Forensics	3(2+1)	
6.	CS461	Parallel & Distributed Computing	3 (3+0)	
Domain Electives (Choose SEVEN Courses)				
1.	CY371	System and Network Administration	3(2+1)	CC261(CN)
2.	CY372	Vulnerability Assessment & Reverse Engineering	3(2+1)	
3.	CY373	Hardware Security	3(2+1)	
4.	CY374	Malware Analysis	3(2+1)	
5.	CY375	Wireless and Mobile Security	3(2+1)	CC261(CN)
6.	CY376	Web App Development and Security	3(2+1)	
7.	CY471	Penetration Testing	3(2+1)	
8.	CY472	Ethical Hacking	3(2+1)	
9.	CY473	Cyber Laws and Crimes	3(2+1)	
10.	<i>Any other programs' elective or domain course subject to the approval of Advisor/Director</i>			

2.12. BSDS PROGRAM DOMAIN AND ELECTIVE COURSES

S#	Code	Course Title	Credit Hrs	Pre-Req.
Domain Core Courses (Select All)				
1.	DS331	Data Science	3 (3+0)	
2.	DS332	Data Warehousing & Business Intelligence	3(2+1)	
3.	DS333	Data Mining	3(2+1)	
4.	DS351	Advanced Statistics	3 (3+0)	QR2
5.	DS361	Data Visualization	3(2+1)	
6.	CS461	Parallel & Distributed Computing	3 (3+0)	
Domain Electives (Choose SEVEN Courses)				
1.	DS371	Business Process Analysis	3(2+1)	
2.	DS372	Platforms & Architectures for Data Science	3 (3+0)	
3.	DS373	Text Mining	3(2+1)	

4.	DS374	Big Data Analytics	3(2+1)	CC221(DB)
5.	AI331	Artificial Neural Networks & Deep Learning	3 (3+0)	
6.	DS472	Data Ethics, Governance and Laws	3 (3+0)	
7.	DS473	Topics in Data Science & AI	3 (3+0)	
8.	<i>All BSAI Domain and Electives courses on priority basis</i>			
9.	<i>Any other programs' elective or domain course subject to the approval of Advisor/Director</i>			

3. PROGRAM'S DETAILS

3.1. PROGRAM LEARNING OUTCOMES FOR COMPUTING PROGRAMS

S#	PLOs	Computing Graduate
1	Academic Education	To prepare graduates as computing professionals
2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements.
3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
6	Individual and Teamwork	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
7	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
8	Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice
9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice
10	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

3.2. PROGRAM EDUCATIONAL OBJECTIVES OF BSCS

S#	Statement
PEO-1	Apply the necessary technical capabilities to contribute effectively to the computing industry.
PEO-2	Manage tasks or research to hone interpersonal and communication abilities through collaboration.
PEO-3	Establish ethical principles, a positive mindset, and societal obligations.

3.3. PROGRAM EDUCATIONAL OBJECTIVES OF BSSE

S#	Statement
PEO-1	Demonstrate strong expertise and attain leadership in software engineering by sound knowledge, skills and abilities.
PEO-2	Implement and manage interpersonal relationships, teamwork, and career development.
PEO-3	Engage in professional practices while taking societal, ethical, and environmental considerations.

3.4. PROGRAM EDUCATIONAL OBJECTIVES OF BSCYS

S#	Statement
PEO-1	Apply cybersecurity principles, tools, and techniques to secure digital assets and contribute effectively to cybersecurity operations and research.
PEO-2	Collaborate in multidisciplinary teams with strong communication and problem-solving skills to manage security challenges
PEO-3	Demonstrate ethical responsibility and consider legal, societal, and global implications in cybersecurity practices.

3.5. PROGRAM EDUCATIONAL OBJECTIVES OF BSAI

S#	Statement
PEO-1	Apply core principles of artificial intelligence to design intelligent systems and contribute effectively to AI-driven industries and research.
PEO-2	Demonstrate teamwork, communication, and leadership skills to manage interdisciplinary projects and adapt to evolving technological environments.
PEO-3	Uphold ethical standards and address societal and environmental impacts while developing responsible AI solutions.

3.6. PROGRAM EDUCATIONAL OBJECTIVES OF BSDS

S#	Statement
PEO-1	Utilize data science methodologies, tools, and analytical techniques to extract insights and provide data-driven solutions across domains.
PEO-2	Work effectively in teams, communicate complex data findings clearly, and adapt to changing technologies and interdisciplinary environments.
PEO-3	Apply ethical principles and address privacy, fairness, and societal impact in data collection, analysis, and decision-making.

3.7. MAPPING PLOS WITH PEOS OF ALL THE PROGRAMS

PLO#	Program Learning Outcome	PEO 1	PEO 2	PEO 3
1	Academic Education	✓		
2	Knowledge for Solving Computing Problems	✓		
3	Problem Analysis	✓		
4	Design / Development of Solutions	✓	✓	
5	Modern Tool Usage		✓	
6	Individual and Teamwork		✓	
7	Communication		✓	✓
8	Computing Professionalism and Society			✓
9	Ethics			✓
10	Lifelong Learning	✓	✓	

3.8. SEMESTER WISE COURSE DISTRIBUTION (GENERIC FRAMEWORK) ■ ■ ■ ■

Sem	Course Title	Credits	Coverage		
Semester 1	 Computing Core (Programming Fundamentals)	4(3+1)	Cat.	Courses	T. Credits
	 General Education (Applications of ICT)	3(3+0)	 GE	4	12
	 General Education (QR-1)	3(3+0)	 Math	1	3
	 Mathematics Course	3(3+0)	 Comp. Core	1	4
	 General Education (Functional English)	3(3+0)	Total	6	19
	 General Education (Islamic Studies)	3(3+0)			
Semester 2	 Computing Core (Object Oriented Prog.)	4(3+1)	Cat.	Courses	T. Credits
	 Computing Core (Digital Logic Design)	3(2+1)	 GE	3	7
	 General Education (Principles of Management)	2(3+0)	 Math	1	3
	 General Education (Intro. to Philosophy)	2(3+0)	 Comp. Core	2	7
	 General Education (Expository Writing)	3(3+0)	Total	6	17
	 Mathematics Course	3(3+0)			
Semester 3	 Computing Core (Data Structures)	4(3+1)	Cat.	Courses	T. Credits
	 Comp. Core (Comp. Org. and Assembly Lang)	3(2+1)	 GE	2	4
	 Computing Core (Software Engineering)	3(3+0)	 Comp. Core	4	14
	 Computing Core (Database Systems)	4(3+1)	Total	6	18
	 General Education (Introductory Mechanics)	3(3+0)			
	 Understanding of Quran - I	1(0+1)			
Semester 4	 Computing Core (Computer Networks)	3(2+1)	Cat.	Courses	T. Credits
	 Computing Core (Operating Systems)	3(3+0)	 GE	5	11
	 Gen. Edu. (Social Ethics and Civil Engagement)	2(2+0)	 Comp. Core	2	6
	 Gen. Edu. (Entrepreneurship)	2(3+0)	Total	7	17
	 Gen. Edu. (Pakistan Studies)	3(3+0)			
	 Gen. Edu. Quantitative Reasoning- II	3(3+0)			
Semester 5	 Computing Core (Artificial Intelligence)	3(2+1)	Cat.	Courses	T. Credits
	 Computing Core (Information Security)	3(3+0)	 Domain Core	2	6
	 Domain Core	3(3+0)	 D. Elective	1	3
	 Domain Core	3(3+0)	 Comp. Core	2	6
	 Domain Elective	3(2+1)	 Social Sc.	1	3
	 Elective Supporting	3(3+0)	Total	6	18
Semester 6	 Domain Core	3(3+0)	Cat.	Courses	T. Credits
	 Domain Core	3(3+0)	 D. Elective	2	6
	 Domain Elective	3(3+0)	 Domain Core	2	6
	 Domain Elective	3(2+1)	 Math	2	6
	 Mathematics (Technical and Business Writing)	3(3+0)	Total	6	18
	 Mathematics	3(3+0)			
Semester 7	 Internship	0	Cat.	Courses	T. Credits
	 Computing Core (Analysis of Algorithms)	3(3+0)	 D. Elective	2	6
	 Domain Elective	3(3+0)	 Domain Core	1	3
	 Domain Elective	3(3+0)	 Comp. Core	2	6
	 Domain Core	3(3+0)	Total	6	15
	 Computing Core (Final Year Project - I)	3(0+3)			
Semester 8	 Computing Core (Final Year Project - II)	3(0+3)	Cat.	Courses	T. Credits
	 Domain Elective	3(3+0)	 D. Elective	2	6
	 Domain Elective	3(3+0)	 GE	1	2
	 Domain Core (Parallel & Dist. Computing)	3(3+0)	 Comp. Core	1	3
	 Ideology and Constitution of Pakistan	2(2+0)	 Domain Core	1	3
			Total	5	14

Total 136 Credit Hours

General Education courses should be scheduled within the first four semesters of the program. The placement of other courses (e.g., Computing Core, Domain Core, Electives) may vary depending on the nature of the program, course prerequisites, and academic requirements. * FYP spans in two semesters and commutatively marked in 8th semester. ** The non-credit course Seerat-un-Nabi (the life and teachings of Prophet Muhammad) be added at any semester as per the feasibility.

3.9. BSCS SEMESTER WISE COURSE DISTRIBUTION

Sem	Course Code	Course Category	Course Title	Credits	Pre-req	
Semester 1	CC102	C.CORE	Programming Fundamentals)	4(3+1)		Total Credits 19
	CS101	GEN	Applications of Information and Communication Technologies	3(3+0)		
	MATH101	GEN	Quantitative Reasoning – I	3(3+0)		
	MATH103	MATH	Discrete Mathematics	3(3+0)		
	ENG 151	GEN	Functional English	3(3+0)		
	RS101	GEN	Islamic Studies	3(3+0)		
Semester 2	CC111	C.CORE	Object Oriented Programming	4(3+1)	CC102 (PF)	Total Credits 17
	CC131	C.CORE	Digital Logic Design	3(2+1)		
	BS311	GEN	Principles of Management	2(3+0)		
	PHI101	GEN	Introduction to Philosophy	2(3+0)		
	ENG152	GEN	Expository Writing	3(3+0)		
	MATH212	MATH	Linear Algebra	3(3+0)		
Semester 3	CC211	C.CORE	Data Structures	4(3+1)	CC102(PF)	Total Credits 18
	CC231	C.CORE	Computer Organization and Assembly Language	3(2+1)	CC131(DLD)	
	CC251	C.CORE	Software Engineering	3(3+0)		
	CC221	C.CORE	Database Systems	4(3+1)		
	PHY101	GEN	Introductory Mechanics	3(3+0)		
	---	GEN	Understanding of Quran - I	1(0+1)		
Semester 4	CC261	C.CORE	Computer Networks	3(2+1)		Total Credits 17
	CC232	C.CORE	Operating Systems)	3(3+0)		
	SW102	GEN	Social Ethics and Civil Engagement	2(2+0)		
	BS311	GEN	Entrepreneurship	2(3+0)		
	PS111	GEN	Pakistan Studies	3(3+0)		
	MATH102	GEN	Quantitative Reasoning – II	3(3+0)		
Semester 5	---	GEN	Understanding of Quran - II	1(0+1)		Total Credits 18
	CC252	C.CORE	Artificial Intelligence	3(2+1)		
	CC262	C.CORE	Information Security	3(3+0)		
	CS311	DOMAIN	Theory of Automata	3(3+0)		
	CS331	DOMAIN	Computer Architecture	3(3+0)	CC131(DLD)	
	---	ELECTIVE	Domain Elective	3(2+1)		
Semester 6	PSY101	SUPPORT	Introduction to Psychology	3(3+0)		Total Credits 18
	CS315	DOMAIN	Compiler Construction	3(3+0)		
	CS321	DOMAIN	Advanced Database Management Systems	3(3+0)	CC221(DB)	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(2+1)		
	ENG334	MATH	Technical and Business Writing	3(3+0)		
Semester 7	MATH272	MATH	Ordinary Differential Equations	3(3+0)		Total Credits 15
	CC497	INTERN	Internship	0		
	CC411	C.CORE	Analysis of Algorithms	3(3+0)	CC211 (DS)	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	CS342	DOMAIN	HCI and Computer Graphics	3(3+0)		
Semester 8	CC498	C.CORE	Final Year Project - I	3(0+3)		Total Credits 14
	CC499	C.CORE	Final Year Project - II	3(0+3)	CC498	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	CS461	DOMAIN	Parallel and Distributed Computing	3(3+0)		
	PS106	GEN	Ideology and Constitution of Pakistan	2(2+0)		

3.10. BSSE SEMESTER WISE COURSE DISTRIBUTION

Sem	Course Code	Course Category	Course Title	Credits	Pre-req	
Semester 1	CC102	C.CORE	Programming Fundamentals)	4(3+1)		Total Credits 19
	CS101	GEN	Applications of Information and Communication Technologies	3(3+0)		
	MATH101	GEN	Quantitative Reasoning-I	3(3+0)		
	MATH103	MATH	Discrete Mathematics	3(3+0)		
	ENG 151	GEN	Functional English	3(3+0)		
	RS101	GEN	Islamic Studies	3(3+0)		
Semester 2	CC111	C.CORE	Object Oriented Programming	4(3+1)	CC102 (PF)	Total Credits 17
	CC131	C.CORE	Digital Logic Design	3(2+1)		
	BS311	GEN	Principles of Management	2(3+0)		
	PHI101	GEN	Introduction to Philosophy	2(3+0)		
	ENG152	GEN	Expository Writing	3(3+0)		
	MATH212	MATH	Linear Algebra	3(3+0)		
Semester 3	CC211	C.CORE	Data Structures	4(3+1)	CC102(PF)	Total Credits 18
	CC231	C.CORE	Computer Organization and Assembly Language	3(2+1)	CC131(DLD)	
	CC251	C.CORE	Software Engineering	3(3+0)		
	CC221	C.CORE	Database Systems	4(3+1)		
	PHY101	GEN	Introductory Mechanics	3(3+0)		
	---	GEN	Understanding of Quran - I	1(0+1)		
Semester 4	CC261	C.CORE	Computer Networks	3(2+1)		Total Credits 17
	CC232	C.CORE	Operating Systems)	3(3+0)		
	SW102	GEN	Social Ethics and Civil Engagement	2(2+0)		
	BS311	GEN	Entrepreneurship	2(3+0)		
	PS111	GEN	Pakistan Studies	3(3+0)		
	MATH102	GEN	Quantitative Reasoning- II	3(3+0)		
Semester 5	---	GEN	Understanding of Quran - II	1(0+1)		Total Credits 18
	CC252	C.CORE	Artificial Intelligence	3(2+1)		
	CC262	C.CORE	Information Security	3(3+0)		
	SE321	DOMAIN	Software Requirement Engineering	3(3+0)		
	SE331	DOMAIN	Software Construction and Development	3(3+0)	CC251(SWE)	
	---	ELECTIVE	Domain Elective	3(2+1)		
Semester 6	PSY101	SUPPORT	Introduction to Psychology	3(3+0)		Total Credits 18
	SE351	DOMAIN	Software Quality Engineering	3(3+0)	CC251(SWE)	
	SE332	DOMAIN	Software Design and Architecture	3(3+0)	CC251(SWE)	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(2+1)		
	ENG334	MATH	Technical and Business Writing	3(3+0)		
Semester 7	MATH272	MATH	Ordinary Differential Equations	3(3+0)		Total Credits 15
	CC497	INTERN	Internship	0		
	CC411	C.CORE	Analysis of Algorithms	3(3+0)	CC211 (DS)	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	SE461	DOMAIN	Software Project Management	3(3+0)		
Semester 8	CC498	C.CORE	Final Year Project - I	3(0+3)		Total Credits 14
	CC499	C.CORE	Final Year Project - II	3(0+3)	CC498	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	CS461	DOMAIN	Parallel and Distributed Computing	3(3+0)		
	PS106	GEN	Ideology and Constitution of Pakistan	2(2+0)		

3.11. BSAI SEMESTER WISE COURSE DISTRIBUTION

Sem	Course Code	Course Category	Course Title	Credits	Pre-req	
Semester 1	CC102	C.CORE	Programming Fundamentals)	4(3+1)		Total Credits 19
	CS101	GEN	Applications of Information and Communication Technologies	3(3+0)		
	MATH101	GEN	Quantitative Reasoning-I	3(3+0)		
	MATH103	MATH	Discrete Mathematics	3(3+0)		
	ENG 151	GEN	Functional English	3(3+0)		
	RS101	GEN	Islamic Studies	3(2+0)		
Semester 2	CC111	C.CORE	Object Oriented Programming	4(3+1)	CC102 (PF)	Total Credits 17
	CC131	C.CORE	Digital Logic Design	3(2+1)		
	BS311	GEN	Principles of Management	2(3+0)		
	PHI101	GEN	Introduction to Philosophy	2(3+0)		
	ENG152	GEN	Expository Writing	3(3+0)		
	MATH212	MATH	Linear Algebra	3(3+0)		
Semester 3	CC211	C.CORE	Data Structures	4(3+1)	CC102(PF)	Total Credits 18
	CC231	C.CORE	Computer Organization and Assembly Language	3(2+1)	CC131(DLD)	
	CC251	C.CORE	Software Engineering	3(3+0)		
	CC221	C.CORE	Database Systems	4(3+1)		
	PHY101	GEN	Introductory Mechanics	3(3+0)		
	---	GEN	Understanding of Quran - I	1(0+1)		
Semester 4	CC261	C.CORE	Computer Networks	3(2+1)		Total Credits 17
	CC232	C.CORE	Operating Systems	3(3+0)		
	SW102	GEN	Social Ethics and Civil Engagement	2(2+0)		
	BS311	GEN	Entrepreneurship	2(3+0)		
	PS111	GEN	Pakistan Studies	3(3+0)		
	MATH102	GEN	Quantitative Reasoning- II	3(3+0)		
Semester 5	---	GEN	Understanding of Quran - II	1(0+1)		Total Credits 18
	CC252	C.CORE	Artificial Intelligence	3(2+1)		
	CC262	C.CORE	Information Security	3(3+0)		
	AI311	DOMAIN	Programming for AI	3(3+0)	CC102(PF)	
	AI322	DOMAIN	Machine Learning	3(3+0)		
	---	ELECTIVE	Domain Elective	3(2+1)		
Semester 6	PSY101	SUPPORT	Introduction to Psychology	3(3+0)		Total Credits 18
	AI331	DOMAIN	Artificial Neural Networks & Deep Learning	3(3+0)		
	AI351	DOMAIN	Knowledge Representation & Reasoning	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(2+1)		
	ENG334	MATH	Technical and Business Writing	3(3+0)		
Semester 7	MATH272	MATH	Ordinary Differential Equations	3(3+0)		Total Credits 15
	CC497	INTERN	Internship	0		
	CC411	C.CORE	Analysis of Algorithms	3(3+0)	CC211 (DS)	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	AI441	DOMAIN	Computer Vision	3(3+0)		
Semester 8	CC498	C.CORE	Final Year Project - I	3(0+3)		Total Credits 14
	CC499	C.CORE	Final Year Project - II	3(0+3)	CC498	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	CS461	DOMAIN	Parallel and Distributed Computing	3(3+0)		
	PS106	GEN	Ideology and Constitution of Pakistan	2(2+0)		

3.12. BSCYS SEMESTER WISE COURSE DISTRIBUTION

Sem	Course Code	Course Category	Course Title	Credits	Pre-req	
Semester 1	CC102	C.CORE	Programming Fundamentals)	4(3+1)		Total Credits 19
	CS101	GEN	Applications of Information and Communication Technologies	3(3+0)		
	MATH101	GEN	Quantitative Reasoning-I	3(3+0)		
	MATH103	MATH	Discrete Mathematics	3(3+0)		
	ENG 151	GEN	Functional English	3(3+0)		
	RS101	GEN	Islamic Studies	3(3+0)		
Semester 2	CC111	C.CORE	Object Oriented Programming	4(3+1)	CC102 (PF)	Total Credits 17
	CC131	C.CORE	Digital Logic Design	3(2+1)		
	BS311	GEN	Principles of Management	2(3+0)		
	PHI101	GEN	Introduction to Philosophy	2(3+0)		
	ENG152	GEN	Expository Writing	3(3+0)		
	MATH212	MATH	Linear Algebra	3(3+0)		
Semester 3	CC211	C.CORE	Data Structures	4(3+1)	CC102(PF)	Total Credits 18
	CC231	C.CORE	Computer Organization and Assembly Language	3(2+1)	CC131(DLD)	
	CC251	C.CORE	Software Engineering	3(3+0)		
	CC221	C.CORE	Database Systems	4(3+1)		
	PHY101	GEN	Introductory Mechanics	3(3+0)		
	---	GEN	Understanding of Quran - I	1(0+1)		
Semester 4	CC261	C.CORE	Computer Networks	3(2+1)		Total Credits 17
	CC232	C.CORE	Operating Systems)	3(3+0)		
	SW102	GEN	Social Ethics and Civil Engagement	2(2+0)		
	BS311	GEN	Entrepreneurship	2(3+0)		
	PS111	GEN	Pakistan Studies	3(3+0)		
	MATH102	GEN	Quantitative Reasoning- II	3(3+0)		
Semester 5	---	GEN	Understanding of Quran - II	1(0+1)		Total Credits 18
	CC252	C.CORE	Artificial Intelligence	3(2+1)		
	CC262	C.CORE	Information Security	3(3+0)		
	CY321	DOMAIN	Cyber Security	3(3+0)		
	CY322	DOMAIN	Network Security	3(3+0)	CC261(CN)	
	PSY101	ELECTIVE	Domain Elective	3(2+1)		
Semester 6	---	SUPPORT	Introduction to Psychology	3(3+0)		Total Credits 18
	CY331	DOMAIN	Secure Software Design and Development	3(3+0)		
	CY351	DOMAIN	Information Assurance	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(2+1)		
	ENG334	MATH	Technical and Business Writing	3(3+0)		
Semester 7	MATH272	MATH	Ordinary Differential Equations	3(3+0)		Total Credits 15
	CC497	INTERN	Internship	0		
	CC411	C.CORE	Analysis of Algorithms	3(3+0)	CC211 (DS)	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	CY441	DOMAIN	Digital Forensics	3(3+0)		
Semester 8	CC498	C.CORE	Final Year Project - I	3(0+3)		Total Credits 14
	CC499	C.CORE	Final Year Project - II	3(0+3)	CC498	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	CS461	DOMAIN	Parallel and Distributed Computing	3(3+0)		
	PS106	GEN	Ideology and Constitution of Pakistan	2(2+0)		

3.13. BSDS SEMESTER WISE COURSE DISTRIBUTION

Sem	Course Code	Course Category	Course Title	Credits	Pre-req	
Semester 1	CC102	C.CORE	Programming Fundamentals)	4(3+1)		Total Credits 19
	CS101	GEN	Applications of Information and Communication Technologies	3(3+0)		
	MATH101	GEN	Quantitative Reasoning-I	3(3+0)		
	MATH103	MATH	Discrete Mathematics	3(3+0)		
	ENG 151	GEN	Functional English	3(3+0)		
	RS101	GEN	Islamic Studies	3(3+0)		
Semester 2	CC111	C.CORE	Object Oriented Programming	4(3+1)	CC102 (PF)	Total Credits 17
	CC131	C.CORE	Digital Logic Design	3(2+1)		
	BS311	GEN	Principles of Management	2(3+0)		
	PHI101	GEN	Introduction to Philosophy	2(3+0)		
	ENG152	GEN	Expository Writing	3(3+0)		
	MATH212	MATH	Linear Algebra	3(3+0)		
Semester 3	CC211	C.CORE	Data Structures	4(3+1)	CC102(PF)	Total Credits 18
	CC231	C.CORE	Computer Organization and Assembly Language	3(2+1)	CC131(DLD)	
	CC251	C.CORE	Software Engineering	3(3+0)		
	CC221	C.CORE	Database Systems	4(3+1)		
	PHY101	GEN	Introductory Mechanics	3(3+0)		
	---	GEN	Understanding of Quran - I	1(0+1)		
Semester 4	CC261	C.CORE	Computer Networks	3(2+1)		Total Credits 17
	CC232	C.CORE	Operating Systems)	3(3+0)		
	SW102	GEN	Social Ethics and Civil Engagement	2(2+0)		
	BS311	GEN	Entrepreneurship	2(3+0)		
	PS111	GEN	Pakistan Studies	3(3+0)		
	MATH102	GEN	Quantitative Reasoning- II	3(3+0)		
Semester 5	---	GEN	Understanding of Quran - II	1(0+1)		Total Credits 18
	CC252	C.CORE	Artificial Intelligence	3(2+1)		
	CC262	C.CORE	Information Security	3(3+0)		
	DS331	DOMAIN	Data Science	3(3+0)		
	DS332	DOMAIN	Data Warehousing & Business Intelligence	3(3+0)		
	---	ELECTIVE	Domain Elective	3(2+1)		
Semester 6	PSY101	SUPPORT	Introduction to Psychology	3(3+0)		Total Credits 18
	DS333	DOMAIN	Data Mining	3(3+0)		
	DS351	DOMAIN	Advanced Statistics	3(3+0)	QR2	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(2+1)		
	ENG334	MATH	Technical and Business Writing	3(3+0)		
Semester 7	MATH272	MATH	Ordinary Differential Equations	3(3+0)		Total Credits 15
	CC497	INTERN	Internship	0		
	CC411	C.CORE	Analysis of Algorithms	3(3+0)	CC211 (DS)	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	DS361	DOMAIN	Data Visualization	3(3+0)		
Semester 8	CC498	C.CORE	Final Year Project - I	3(0+3)		Total Credits 14
	CC499	C.CORE	Final Year Project - II	3(0+3)	CC498	
	---	ELECTIVE	Domain Elective	3(3+0)		
	---	ELECTIVE	Domain Elective	3(3+0)		
	CS461	DOMAIN	Parallel and Distributed Computing	3(3+0)		
	PS106	GEN	Ideology and Constitution of Pakistan	2(2+0)		